Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI3371-WE01

Title: MEng and BEng Engineering

Electrical Engineering 3

Time Allowed:	2 hours	
Additional Material provided:	None.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries: YES		

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each section. All relevant workings must be shown.

Revision:	

Section A

Question 1

Figure Q1 shows a DC motor drive system using a back-to-back arrangement of two single phase full-wave thyristor converters. The converters are fed from a single-phase AC source with an RMS voltage of 230 V at 50 Hz. The separately excited DC motor has an armature resistance of 0.05 Ω and the motor constant (K) is 0.2 V/RPM. The motor is under constant field operation and the converters are assumed to be loss-less. The drive is capable of four-quadrant operation.

a) Sketch a typical output voltage waveform of a single phase full-wave thyristor converter and derive an expression for its average output DC voltage in terms of the RMS value of the AC single phase source voltage and the firing angle of the converter.

[40%]

b) Under normal conditions, the motor in Figure Q1 is driving a full load torque of 250 Nm at 850 RPM in forward motion with converter $C1$ conducting.

i. Calculate the required firing angle for converter $C1$ to drive the machine at full load. Meanwhile, calculate the efficiency of the motor under full load operation. Neglect any friction and windage (mechanical) losses.

[20%]

ii. Under braking conditions from forward motoring, calculate the required firing angle for the conducting converter in order to maintain the braking torque at 75% of the above full load torque. Assume that the mass of the armature is large enough to prevent any instantaneous changes in the motor speed during the initial stages of the braking.

[30%]

iii. Meanwhile, calculate the total energy (in kilowatt-seconds) recovered by the source during the braking operation if the machine were to stop in 3 seconds.

[10%]

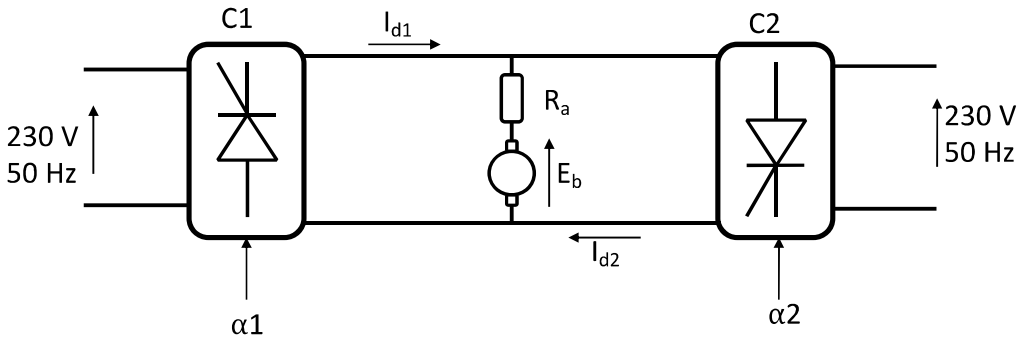


Figure Q1 – A DC Motor Drive with Two Back-to-Back Full-Wave Thyristor Converters

Question 2

A 4-pole, three-phase, star-connected induction motor is being driven by a three-phase PWM-controlled Voltage Sourced Inverter (VSI). The motor is under volt/hertz control and it is initially started at 600 V/50 Hz (RMS, line-line) driving a centrifugal pump in which the load torque is proportional to the square of the motor speed. The per-phase equivalent circuit parameters are shown in Table Q2.

a) Aided by sketches of the appropriate torque-speed regions (quadrants) of operation qualitatively describe the trajectory of the operating point if the supply is suddenly changed to 300 V/25 Hz of the same phase sequence.

[30%]

b) Using the simplified per-phase equivalent circuit of the induction motor, calculate the direct on-line current drawn by the motor when at starting and when at full load. Assume the motor supply is at the initial value of 600 V/ 50 Hz (RMS, line-line) and at full load it is running at 1330 RPM.

[30%]

c) Determine the rotor speed at which the motor develops its maximum torque for when it is supplied at 600 V/50 Hz (RMS, line-line) and when it is at 300 V/25 Hz (RMS, line-line).

[20%]

d) Determine the maximum torque output by the machine when operating at 600 V/50 Hz (RMS, line-line) and when operating at 300 V/25 Hz (RMS, line-line). Draw any conclusions you can make based on your comparison of results of maximum torque outputs about the reasons of the motor behaviour.

[20%]

Table Q2 – Induction Motor Per-phase Equivalent Circuit Parameters
(All values referred to the Stator)

Parameter	Stator Winding	Rotor Winding (referred to stator)	Magnetising (Core) Branch
Resistance	0.4 Ω	1.01 Ω	200 Ω
Reactance	1.7 Ω	0.95 Ω	100 Ω

Section B

Question 3

a)

Consider a medium length transmission line where three of the generalised circuit constants (A, B, D) are given as below:

$$A = D = 0.936 + j0.016, \qquad B = 33.5 + j138$$

i. Find constant C.
ii. The load at the receiving end is 35 MW at 220 kV with a power factor of 0.9 lagging. Find the magnitude of the sending end voltage and the voltage regulation.
iii. Consider another case where the load at the receiving end is 40 MW with a power factor of 0.9 lagging, and the sending end voltage is 225 kV. Find the magnitude of the receiving end voltage in this case.

[70%]

b)

In power systems, sometimes voltages, currents and powers are expressed in per-unit, or as per unit of a base or reference value.

i. Explain how the per-unit phase and line voltages are related to each other and why.

ii. Meanwhile, explain three advantages of using the per-unit system in power system analysis.

[10%]

c)

Deloading wind turbines means that a wind power plant is required to feed in less than its available active power, which allows the wind power plant to ramp power up or down in response to system frequency variations. The two primary methods for deloading are overspeeding and pitching. Explain in detail what these methods are, and how they are selected based on wind speed.

[10%]

d)

Describe how the reserve generation requirement for a power system including wind energy generation is calculated.

[10%]

Question 4

a)

Consider a power system where three generators are running in parallel. Generator 1 has a nominal rated power $P_{g1,r} = 200\,MW$ and its speed governor has a droop $R_{g1} = 0.02\,pu$. Generator 2 has a nominal rated power $P_{g2,r} = 200\,MW$ and its speed governor has a droop $R_{g2} = 0.06\,pu$. Generator 3 has a nominal rated power $P_{g3,r} = 200\,MW$ and its speed governor has a droop $R_{g2} = 0.06\,pu$.

Assume the load suddenly consumes $20\,MW$ more active power.

i. Determine how the active power increase of the load will be distributed over the three generators.

ii. Calculate how much the power system frequency will change after this increase of the load.

[30%]

b)

i. Describe what a Thyristor-Controlled Reactor (TCR) is, and how it works. Explain how a TCR can be used to improve a power system's operation.

ii. One of power quality problems in a power system is presence of current harmonics consumed by nonlinear loads such as power electronics-based ones. A typical solution for this problem is to use harmonic filters such as active power filters. Explain what an active power filter is, how it works, and how it can be used to solve harmonic current problem.

[10%]

c)

Consider the two-bus system shown in Figure Q4, where bus 1 is a slack bus with the voltage of $V_1 = 1.0\angle 0°$ per unit (pu). A load of 100 MW and 20 MVAr is taken from bus 2. The two buses are connected by a transmission line with admittance $y_{12} = a + jb$ pu on a base of 100 MVA, where $a = 0.5882$ and b is unknown. Using Fast-Decoupled method, and after one iteration, the new estimate for the voltage phase angle at bus 2 is $\delta_2^{(1)} = -0.425$ radian.

Calculate b. Also obtain the voltage magnitude of bus 2 after the first iteration. Use an initial estimate of $|V_2|^{(0)} = 1.0$ pu and $\delta_2^{(0)} = 0°$.

[60%]

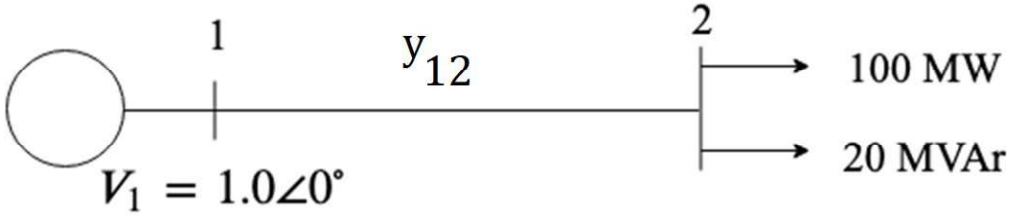


Figure Q4 network diagram

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